

MANIDEEP EMMADI

Hyderabad, India

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Education

B.Tech in Electronics and Communication Engineering **2024 – 2027**
Methodist college of Engineering and Technology *India*

Diploma in Electronics and Communication Engineering **2021 – 2024**
Mahaveer Institute of Science and Technology *India*

Relevant Coursework

- Digital Electronics
- Signals and Systems
- Microprocessors
- Semiconductor Physics
- Analog Electronics
- VLSI Design
- Embedded Systems
- Control Systems

Experience

DRDO – Research Centre Imarat (RCI) **2025**
Electronics / VLSI Intern *Hyderabad, India*

- Gained exposure to digital design concepts and hardware verification techniques.
- Worked with Verilog testbench structures and simulation environments.
- Observed semiconductor design workflows used in defense electronics systems.

Techieyaan

Internship Trainee **2023**
Hyderabad

- Completed internship training focused on electronics and embedded systems fundamentals.
- Worked on small technical assignments related to IoT and hardware interfacing.
- Gained hands-on exposure to Arduino based systems and basic circuit design.
- Collaborated with peers on technical learning modules and mini projects.

Projects

16-bit ALU Processor Design | *Digital Design, Circuit Simulation*

- Designed a 16-bit Arithmetic Logic Unit capable of performing multiple arithmetic and logical operations.
- Performed power analysis and transient analysis to evaluate circuit performance.
- Analyzed timing behavior and efficiency of the processor components.

Traffic Light Controller FSM | *Verilog, Xilinx Vivado*

- Designed a finite state machine (FSM) based traffic light controller using Verilog HDL.
- Simulated and verified system functionality using Xilinx Vivado tools.
- Implemented optimized state transitions for traffic signal control.

IoT Urban Drainage Monitoring System | *Arduino, IoT Sensors*

- Developed an IoT-based monitoring system to track drainage water levels in urban environments.
- Integrated sensors with Arduino for real-time monitoring and alerts.
- Designed the system to support smart city infrastructure and flood prevention.

Electrostatics Simulation | *Python, NumPy, Matplotlib*

- Created a Python simulation to visualize electric field distribution using numerical methods.
- Implemented data visualization using NumPy and Matplotlib libraries.

Technical Skills

Programming Languages: Verilog, Python, C
Hardware / Embedded Platforms: Arduino, Sensor Interfacing
EDA & Development Tools: Xilinx Vivado, Arduino IDE, VS Code, Custom Compiler
Simulation & Analysis: Circuit Simulation, Power Analysis, Transient Analysis
Libraries: NumPy, Matplotlib
Domains: VLSI Design, Digital Electronics, Embedded Systems, IoT Systems

VLSI & Hardware Skills

HDL: Verilog
Digital Design Concepts: Finite State Machines (FSM), ALU Design, Digital Logic
EDA Tools: Xilinx Vivado
Circuit Analysis: Power Analysis, Transient Analysis
Hardware Platforms: Arduino, Sensor-based Systems

Leadership

Street Cause	2024 – Present
<i>Divisional President</i>	

- Led divisional initiatives and coordinated volunteers for community service activities.
- Successfully raised over 10,000 as part of the RFC 11 fundraising campaign.
- Organized and managed social impact programs and outreach events.